

Distributed LinUCB-Based Adaptive DB-LBT with Local Observations for Wi-Fi/NR-U Coexistence

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Abstract—Deterministic backoff with listen before talk (DB-LBT) uses local countdown interruptions to build collision-reduced Wi-Fi/NR-U access schedules. Its recovery parameters are normally fixed, although offered load, active contenders, and external occupancy change the backoff profile that works best. We formulate DB-LBT parameter adaptation as a protocol-constrained contextual bandit problem under locally observable MAC information. Each transmitter runs LinUCB over 24 legal $(\kappa, \beta, m, b_{\text{init}})$ tuples. The context contains only local CCA, queue, delay, retransmission, and ACK/HARQ statistics; the selected action changes the next DB-LBT recovery counter but leaves native Wi-Fi and NR-U procedures intact.

We derive the sensing, reception, separability, and timescale conditions that make this local context informative. Paired event experiments then compare adaptive DB-LBT with random LBT and fixed DB-LBT. In a static $4 + 4$ regime, adaptive and fixed TMC DB-LBT are numerically identical. Across held-out scenarios, adaptation raises mean utility from 0.926467 to 0.928023; the non-ideal gain is 0.289%. LinUCB also exceeds context-free UCB in the combined dynamic case. A packet-level ns-3 cross-check produces mixed technology-level directions and exposes the unresolved validation gap.

Index Terms—Wi-Fi, NR-U, coexistence, listen before talk, deterministic backoff, contextual bandit, local observation.

I. INTRODUCTION

Wi-Fi and 5G New Radio in unlicensed spectrum (NR-U) contend without a common scheduler. Both technologies defer to detected energy and count down a random backoff, but the surrounding procedures differ. Wi-Fi follows DCF/EDCA after an inter-frame deferral. NR-U category-4 LBT must also obey channel-occupancy limits and align access to an NR transmission boundary [1]–[3]. When several counters expire together, collisions waste airtime and extend delay tails.

DB-LBT replaces part of this repeated random recovery with a counter derived from local countdown interruptions [4]. The resulting spacing can approach a collision-free access order without node identities or cross-technology messages. Later work showed that the mechanism also works under non-full-buffer traffic, random busy slots, sensing perturbation, and asymmetric energy detection [5]. Those regimes also expose a practical weakness: one fixed recovery tuple must serve both a quiet cell and a cell whose active set or queues have changed.

Consider a node that has just completed an attempt. It knows whether CCA froze its countdown, how long packets waited, whether ACK/HARQ succeeded, and how its queue evolved. It does not know every peer queue or the instantaneous channel

gain. The useful control question is which legal DB-LBT recovery profile should be used next. We formulate DB-LBT parameter adaptation as a protocol-constrained contextual bandit problem under locally observable MAC information.

This formulation targets fixed indoor cells with mutual sensing and coarse changes in offered load, active contenders, or persistent occupancy. Within a phase, local histories must remain stable long enough to guide several updates. Across phases, the same histories must change enough to distinguish profiles. Such traffic and occupancy structure is common in scheduled application bursts, shifted active sets, and periodic incumbent activity even though individual CCA samples are random.

The contributions are threefold. First, we define a 24-arm contextual bandit whose actions are legal DB-LBT tuples; learning cannot alter sensing thresholds, Wi-Fi deferral, NR-U synchronization or occupancy limits, PHY rate, or retransmission semantics. Second, we derive conditions under which local CCA/MAC history provides an actionable context: the contention abstraction must hold, regimes that require different actions must be locally separable, and their dwell time must exceed observation and adaptation windows. Third, paired event experiments show that adaptation preserves the stable DB-LBT regime and provides a small held-out gain, while a packet-level cross-check exposes a mismatch for future experiments. Fast radio variation and strategic co-adaptation are treated as scope boundaries in Section VI.

II. RELATED WORK

Wi-Fi/NR-U coexistence schemes tune contention windows, reserve or defer airtime, resolve mini-slot collisions, and schedule around technology-specific timing [3], [6], [7]. Deterministic and semi-random WLAN backoff instead creates post-success structure without explicit slot assignments [8], [9]. DB-LBT extends this principle across Wi-Fi and NR-U: countdown interruptions estimate active contention, and a deterministic recovery counter spaces later attempts [4]. Its TMC extension studies convergence with periodic schedules and random busy slots, then evaluates traffic, interference, and sensing perturbations [5]. Both provide the fixed mechanism on which our adaptive action space is built.

Machine learning has been applied throughout the IEEE 802.11 stack [10]. LTE-LAA work adapts contention windows

from observed performance [11], while reinforcement learning has optimized technology-level efficiency and fairness [12]. Our setting is smaller and more constrained. A transmitter chooses among 24 protocol-valid recovery profiles, observes only its own outcome, and must learn with limited online samples. These properties favor a contextual bandit with an explicit uncertainty bonus [13]. The contribution is the mapping from local protocol measurements to this constrained decision problem, rather than a new generic bandit algorithm.

III. LOCAL INFORMATION AND PROTOCOL ABSTRACTION

A. Where Adaptation Enters

Let $\mathcal{K} = \mathcal{K}_W \cup \mathcal{K}_N$ contain the Wi-Fi APs and NR-U gNBs sharing one channel. Wi-Fi decrements a counter only after the applicable inter-frame deferral. NR-U category-4 LBT counts idle sensing slots and must also respect its priority class, maximum channel occupancy, and transmission-boundary alignment [2], [7]. DB-LBT acts after an ACK/HARQ outcome: it supplies the next recovery counter before either native access procedure starts.

For a local countdown slot ℓ , $c_k(\ell)$ evolves as

$$c_k(\ell + 1) = \begin{cases} c_k(\ell) - 1, & C_k(\ell) = 0, \\ c_k(\ell), & C_k(\ell) = 1, \end{cases} \quad (1)$$

where $C_k = 1$ denotes a busy CCA sample. A maximal busy episode freezes the counter and increments the local interruption count i_k . In a mutually detectable collision domain, each successful peer contributes roughly one such episode per cycle, giving the operational estimate $\hat{n}_k \approx 1 + i_k$.

B. From the Channel to Local Context

The received power from transmitter j includes path loss, shadowing, and small-scale fading through gain $G_{jk}(t)$. LBT compresses their sum into a threshold decision, while the receiver compresses SINR γ_k into an ACK/HARQ outcome:

$$C_k(t) = \mathbb{1} \left\{ N_k + \sum_{j \neq k} a_j(t) P_j G_{jk}(t) \geq \Gamma_k \right\}, \quad (2)$$

$$\Pr\{Y_k = 1 \mid \gamma_k, u_k\} = 1 - \text{PER}_{\tau_k}(\gamma_k, u_k). \quad (3)$$

Both mappings are nonlinear. A small received-power change can flip CCA, and a failed attempt can reflect overlap, fading, hidden interference, or decoding. The learner receives protocol outcomes, not the underlying gains.

Three assumptions recover the contention abstraction used by DB-LBT: (A1) every co-channel transmission exceeds every contender's CCA threshold; (A2) isolated reception succeeds with probability near one; and (A3) simultaneous transmissions have no capture. Under A1–A3,

$$C_k(\ell) = \mathbb{1} \left\{ \sum_{j \neq k} a_j(\ell) > 0 \right\}, \quad (4)$$

$$Y_k(\ell) = 1 \iff \sum_j a_j(\ell) = 1.$$

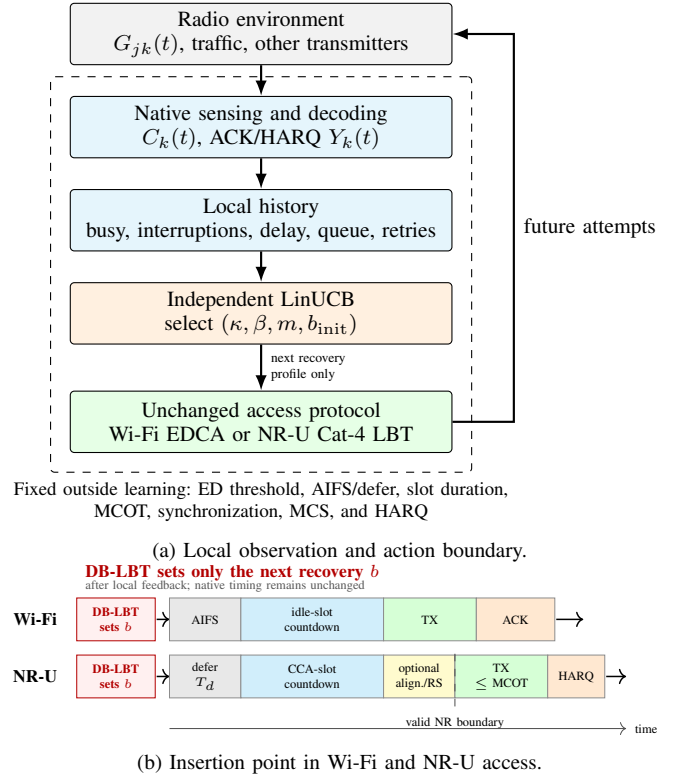


Fig. 1. Protocol-constrained adaptation. LinUCB selects only the next DB-LBT recovery profile from local history. Native sensing, countdown, synchronization, MCOT, and ACK/HARQ behavior remain unchanged.

TABLE I
LOCAL CONTEXT AS PROTOCOL EVIDENCE

Feature group	Local protocol source	Regime cue
Busy fraction, interruptions	CCA thresholding and countdown freezes	Active contenders or persistent occupancy.
Success, failures, retries	ACK/HARQ and retransmission state	Service quality after attempted access.
Queue, arrivals, delay	Local traffic and MAC service	Offered load and congestion.
Effective-data time, busy share	Successful occupancy and NR-U alignment	Schedule efficiency and local share.

One zero counter then succeeds and multiple zeros collide. This reduction explains why busy periods and interruptions carry contention information even though the radio channel itself is random.

The context groups 11 quantities already available at an AP or gNB. Busy fraction and interruptions summarize C_k ; success, failure, and retransmissions summarize Y_k ; queue occupancy, arrivals, delay, and effective-data time describe demand and service.

The feature groups respond at different moments in an access cycle. When several idle queues become active together, the first attempts expose the activation range through queue growth, delay, and early CCA freezes. Once the active set settles, repeated interruptions and retransmissions describe how well the deterministic and random recovery branches maintain spacing. A periodic external transmitter creates an

TABLE II
CONTEXTUAL-BANDIT ACTIONS AND PROTOCOL MEANING

Variable	Values	DB-LBT role
κ	5, 7	Recovery-cycle length indexed by retransmissions.
β	2, 3	Early states using interruption-derived spacing.
m	4, 6, 10	Maximum random backoff after repeated failure.
b_{init}	15, 31	Initial range when a queue becomes active.
α	fixed 11	Common idle-space offset.

other pattern: busy fraction rises at regular intervals even when the peer population is unchanged. The controller does not have to label these causes. It only needs their local traces to rank the available recovery profiles.

Let z denote a MAC occupancy regime and $\bar{x}_k(z)$ its mean local context. Table I shows why each feature group can change when load, active set, or external occupancy changes. Local information separates the actions that matter when

$$q^*(z) \neq q^*(z') \implies \|\bar{x}_k(z) - \bar{x}_k(z')\|_2 \geq \delta_x > 0. \quad (5)$$

Timing supplies the second condition. Five scales describe the operating region: T_{phy} for fast radio variation, T_{ctx} for context collection, T_{adapt} for one learning response, T_{dwell} for a regime, and T_{hor} for the operating horizon. The intended setting satisfies

$$T_{\text{phy}} \ll T_{\text{ctx}} \lesssim T_{\text{adapt}} \ll T_{\text{dwell}} < T_{\text{hor}}. \quad (6)$$

Fast randomness can then average into local history before the regime changes, while at least one change occurs soon enough to make adaptation useful. Equations (5) and (6) give local observations their precise role: they need only distinguish DB-LBT actions, not reconstruct the physical channel.

IV. PROTOCOL-CONSTRAINED CONTEXTUAL BANDIT

A. Legal DB-LBT Actions

After queue activation, node k draws $b_k \sim \mathcal{U}[0, b_{\text{init}}]$. Once an attempt completes, DB-LBT sets the recovery counter

$$b_k^+ = \begin{cases} \alpha + i_k, & r_k \bmod \kappa < \beta, \\ \mathcal{U}[0, m], & \text{otherwise,} \end{cases} \quad (7)$$

where r_k is the retransmission count. The deterministic branch turns interruptions into schedule spacing; the short random branch separates coincident schedules. We retain $\alpha = 11$ and expose only the four recovery parameters in Table II.

The Cartesian product contains 24 arms and always satisfies $\beta < \kappa$. It is centered on the published TMC tuple (7, 3, 6, 15): $b_{\text{init}} = 15$ matches the common minimum contention window, 31 gives a more conservative activation range, and each m is below α . An action supplies only the next counter. CCA thresholds, AIFS/defer timing, NR-U priority class and MCOT, synchronization, MCS, and ACK/HARQ processing stay fixed by the technology and scenario.

Each parameter addresses a distinct recovery moment. The activation range b_{init} controls how aggressively a newly

nonempty queue enters an existing schedule. Parameters κ and β determine how often early retransmission states return to interruption-derived spacing, while m controls the random escape after repeated failure. A larger value is not uniformly better: conservative activation can avoid a burst collision but adds delay when few nodes are active, and extra random recovery can separate colliding schedules while weakening deterministic reuse. This tradeoff is the reason to select complete tuples rather than tune one scalar.

B. Local LinUCB Controller

Each node retains at most 64 of its own attempts. Its 11-dimensional context x_k contains success/failure ratios, CCA busy fraction, countdown interruptions, delay EWMA and P95, queue occupancy, arrivals, retransmissions, effective-data occupancy, and $\hat{n}_k = \text{clip}(1 + \hat{i}_k, 1, 32)$. No peer identity, peer queue, global fairness value, or cross-technology message enters the vector.

The action space is discrete, the context is low dimensional, and online samples are scarce. LinUCB fits this setting with one linear reward model per arm and an explicit exploration bonus. Within a locally separable regime, the working approximation is

$$\mathbb{E}[R_{k,t} \mid x_{k,t} = x, q_{k,t} = q, z] \simeq x^\top \theta_{k,q}^{(z)}. \quad (8)$$

The model is linear in context within one arm; it is not linear in the four DB-LBT parameters. Keeping a separate $\theta_{k,q}^{(z)}$ for every tuple allows parameter interactions to differ across arms. It also avoids imposing a monotone ordering on κ , β , m , or b_{init} . Thus the nonlinear CCA and decoding maps in (2)–(3) remain outside the learner, while the locally observed reward surface is approximated where decisions are actually made.

For each arm q , the controller maintains $A_q \in \mathbb{R}^{11 \times 11}$ and $d_q \in \mathbb{R}^{11}$:

$$\hat{\theta}_q = A_q^{-1} d_q, \quad (9)$$

$$p_q(x_k) = \hat{\theta}_q^\top x_k + c \sqrt{x_k^\top A_q^{-1} x_k}, \quad (10)$$

$$q^* = \arg \max_q p_q(x_k), \quad (11)$$

with $c = 0.5$. After observing reward R_k , the selected model receives $A_q \leftarrow A_q + x_k x_k^\top$ and $d_q \leftarrow d_q + R_k x_k$.

The reward is local as well. Let A_k be effective-airtime fraction, $D_{95,k}$ P95 delay, s_k the node's successful-busy share, and $P_{f,k}$ its failure ratio:

$$U_A = \text{clip}(\hat{n}_k A_k, 0, 1), \quad (12)$$

$$U_D = 1 - \text{clip}\left(\frac{D_{95,k}}{16 \text{ ms } \hat{n}_k}, 0, 1\right), \quad (13)$$

$$U_S = 1 - \text{clip}\left(\frac{|s_k - 1/\hat{n}_k|}{1/\hat{n}_k}, 0, 1\right), \quad (14)$$

$$R_k = (U_A + U_D + U_S)/3 - 0.25 P_{f,k}. \quad (15)$$

Every transmitter owns a controller initialized from the same offline model. After eight observations, the event implementation decides every 32 global contention rounds; the packet implementation decides every 32 completed local attempts. A

decision affects the next recovery only, never a countdown already in progress.

C. Initialization and Online Sequence

Offline initialization evaluates every arm under 11 topology, load, interference, and sensing conditions using three training seeds. Those samples populate the 24 linear models before formal evaluation; none of the ten formal event seeds is used for initialization or fixed-oracle selection. The same initial model is copied to each transmitter, but subsequent contexts, choices, rewards, and updates are local.

An online step follows the protocol clock. The node first aggregates its latest history into x_k , selects q^* with (11), and applies that tuple only when the current attempt has completed. It then observes the next reward window and updates the chosen arm. Separating observation, selection, application, and reward assignment prevents an action from receiving credit for a countdown that began under an earlier profile.

V. EVALUATION

The evaluation reuses the completed paired-seed data. The event simulator implements (4) for 100,000 contention rounds, with 2 ms transmissions, equal Wi-Fi/NR-U contention windows [15, 63], and a 250 μ s NR-U synchronization grid. Ten formal seeds are disjoint from the three pretraining seeds. Pairing fixes exogenous random streams across policies. Held-out conditions cover unseen symmetric and asymmetric densities, two Poisson rates, two periodic occupancies, sensing perturbation, and a combined dynamic case. A separate ns-3.35/5G-LENA/NR-U model provides the packet-level cross-check.

The reporting utility is distinct from the local reward in (15):

$$U_D^{\text{eval}} = 1 - \text{clip}(D_{95}/500 \text{ ms}, 0, 1), \quad (16)$$

$$U_{\text{eval}} = (A_{\text{total}} + U_D^{\text{eval}} + J)/3 - 0.25P_{\text{col}}. \quad (17)$$

It combines effective data airtime, P95 access delay, Jain fairness, and event collision probability. Full scenario matrices, seeds, and component metrics appear in the appendix.

The experiments follow the control argument in order. Mechanism validation checks that online selection does not erase deterministic spacing in a stable cell. Held-out event scenarios then ask whether the pretrained controller transfers to unseen density, asymmetry, traffic, and occupancy conditions. The packet model replaces the ideal contention rule with EDCA, HARQ, PHY interference, and NR-U alignment. Finally, ablations compare contextual and context-free selection and remove feature groups from the same dynamic case. This sequence keeps a scheduling result, a generalization result, and a packet-level result from being treated as interchangeable evidence.

A. Mechanism Validation

Table III first asks whether adding a learner disturbs the underlying schedule. In static 4 + 4, deterministic spacing removes almost all event collisions and the random-backoff

TABLE III
MECHANISM VALIDATION OVER TEN EVENT SEEDS

Scenario	Policy	U_{eval}	P_{col}	D_{95} (ms)
Static 4 + 4	Random	0.805443	0.229707	77.881
	Fixed TMC	0.979499	0.000139	14.018
	Adaptive	0.979499	0.000139	14.018
Load-change 4 + 4	Random	0.814530	0.174006	93.110
	Fixed TMC	0.961087	0.017734	14.018
	Adaptive	0.961785	0.016664	14.018

delay tail. Adaptive and fixed TMC DB-LBT are numerically identical. After a load change, the adaptive controller retains the mechanism and improves utility from 0.961087 to 0.961785.

B. Generalization beyond Training Conditions

Figure 2(a) aggregates ten unseen topology, load, and occupancy scenarios. Adaptive DB-LBT reaches 0.928023, compared with 0.926467 for fixed TMC and 0.928067 for the training-seed fixed oracle. The paired gain is 0.001556, with 95% interval $[0.001470, 0.001651]$. Stable utility changes by only -7.0×10^{-7} , and Jain fairness stays within the registered -0.01 tolerance.

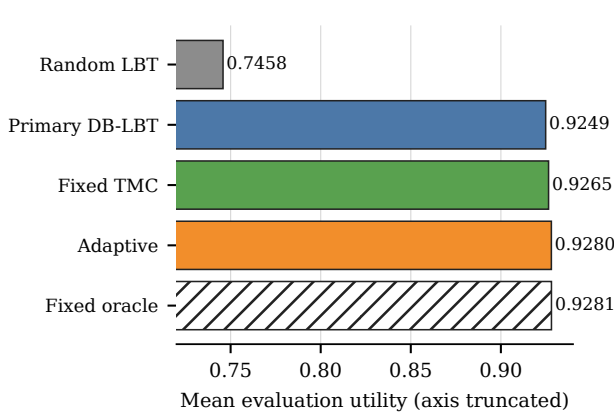
Each confidence interval is formed from ten within-seed policy differences, not from pooled rounds. The exogenous seed controls arrivals, occupancy, and other scenario randomness shared by the compared policies. The fixed oracle is chosen only from training seeds and is included as a reference for a single non-adaptive tuple, not as an online upper bound.

The aggregate hides useful variation. Light Poisson traffic and the combined dynamic case improve, heavy Poisson traffic slightly degrades, and several periodic or sensing-perturbation cases tie at retained precision [Fig. 2(b)]. Across the registered non-ideal set, the gain is 0.289%, below the 10% engineering target. The adaptive controller is therefore most useful where local load and active-set information changes the preferred recovery profile.

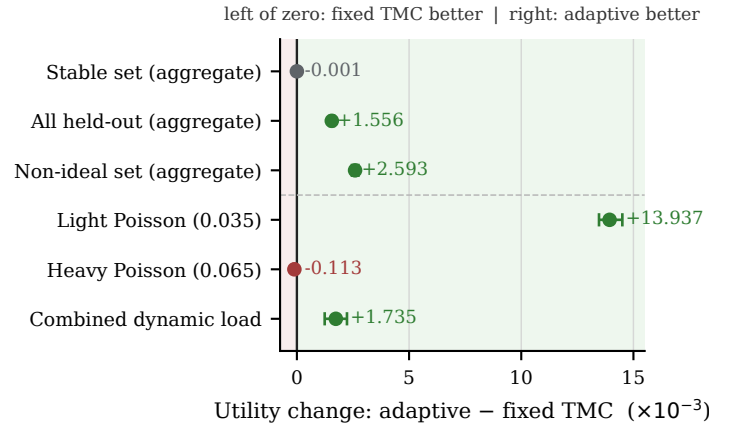
C. Packet-Level Cross-Check

The ns-3 model retains Wi-Fi ACK/EDCA behavior, NR-U HARQ and boundary alignment, a 3GPP indoor frequency-selective channel, and a periodic waveform interferer. Node positions and the sampled channel matrix are fixed. The load-change case switches offered load halfway through its two-second run, making it the closest packet-level realization of the intended phase change.

Across three paired seeds, adaptive-minus-fixed changes mix favorable and unfavorable Wi-Fi/NR-U throughput and delay components. The event direction is tied in the static and periodic cases; for the load change it is favorable, whereas the packet-level majority is unfavorable. The stored ns-3 field named `collision_probability` is reported as packet-loss ratio because it is computed from transmitted minus received IP packets and can include decoding failure. The appendix gives every technology-level difference.



(a) Mean utility across held-out scenarios.



(b) Paired adaptive-minus-fixed-TMC differences.

Fig. 2. Held-out event evaluation over ten paired seeds. In (a), the axis is truncated and values are printed beside the bars. In (b), differences are scaled by 10^3 ; bars are 95% paired-bootstrap intervals. Green favors adaptive DB-LBT, red favors fixed TMC, and gray crosses zero.

D. What the Context Adds

The combined dynamic event case isolates the controller choices. Replacing LinUCB with context-free UCB lowers utility by 0.000686, with 95% interval $[-0.001259, -0.000138]$. This is the clearest evidence that the local feature vector matters. Removing queue, delay, or CCA/interruption features one group at a time yields intervals that cross zero, as does freezing online updates. The complete ablation forest is retained in the appendix.

VI. DISCUSSION

A. Why a Linear Contextual Model Is Enough

The learner does not predict a waveform or an instantaneous channel gain. Native CCA and ACK/HARQ processing have already converted those quantities into busy periods, interruptions, successes, and delays. For the controller, the relevant surface is the reward difference among 24 legal recovery profiles. An 11-dimensional linear model is a useful first approximation to that small surface, and LinUCB adds directed exploration when online samples are limited. The UCB ablation supports the value of context; the unresolved feature ablations do not motivate a larger nonlinear model.

The scenario differences also fit the protocol interpretation. Under light Poisson traffic, queues repeatedly enter and leave contention, so activation and recent service history can distinguish useful profiles. Heavy Poisson traffic behaves more like a persistently active set, where fixed TMC is already strong and exploration can cost utility. The combined case mixes arrivals, periodic occupancy, and active-set turnover; there, contextual LinUCB separates from UCB even though no single feature-group removal is decisive. These patterns explain the aggregate gain without suggesting that adaptation should improve every load.

B. Distributed-Learner Interpretation

When several nodes adapt, node k receives reward $R_k(\mathbf{q}_t)$ under the joint arm vector $\mathbf{q}_t = (q_{1,t}, \dots, q_{|\mathcal{K}|,t})$. This in-

teraction may be read as a restricted stochastic contextual bandit game. The reading is valid when peer actions are non-anticipating within the current decision interval, their joint policy changes little over T_{adapt} , and updates are asynchronous or weakly coupled. Peers may learn under those conditions because their aggregate behavior changes slowly enough over a local update window. Rapid synchronized co-adaptation, action-conditioned retaliation, or jamming breaks that approximation. The interpretation supplies neither a Nash convergence result nor an adversarial-regret guarantee.

C. Limitations and Future Work

The operating region lies between two unhelpful extremes. A fixed occupancy process needs no adaptation; a process that changes faster than context can be collected makes history stale. Equation (6) instead assumes locally distinguishable phases with enough dwell time for learning. Separate Poisson-rate runs establish load dependence, while the event dynamic-combined-4x4 case is a faster stress test: nodes join every 10 global rounds, stay active for 200 rounds, and decisions occur every 32 rounds. Neither experiment directly estimates T_{adapt} .

Physical validation is also incomplete. The ns-3 runs use fixed positions, disabled log-normal shadowing, one static frequency-selective channel draw, and only three two-second seeds. A stronger study should separate propagation and decoding measurements (SINR, PER/BLER, MCS, coherence time) from sensing measurements (energy-detection margin and miss/false-busy rates). MAC service needs simultaneous-access collisions, airtime, retransmissions, and delay tails; learning dynamics needs recovery time and performance versus $T_{\text{adapt}}/T_{\text{dwell}}$. These four measurement classes would show whether a sliding-window learner, change-point reset, or nonlinear reward model is actually warranted.

VII. CONCLUSION

We formulated DB-LBT parameter adaptation as a protocol-constrained contextual bandit driven by local MAC information. Under mutual sensing, reliable isolated reception, local separability, and sufficient regime dwell time, the 11 measured features distinguish when a different recovery tuple can help. Distributed LinUCB then selects among 24 legal profiles without changing native Wi-Fi or NR-U access rules. The completed event study preserves the stable DB-LBT regime and finds a small held-out gain, with the clearest contextual benefit under combined dynamic load. Packet-level directions are mixed. Longer ns-3 campaigns with controlled fading, sensing, and adaptation timescales are the next step.

APPENDIX

The complete reproducibility package, including source code, configurations, seeds, results, and figures, is available at:

<https://github.com/Ignister01/distributed-linucb-dblbt>

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